

README

ota-update-engine-bridge

Field	Value
Revision	2
Created	2026-06-07
Status	implemented (:core fully unit-tested; :android real, AGP — see BUILD_STATUS.md)
Part of	Helix OTA
Gradle modules	:core (Kotlin/JVM), :android (Android library)
Language	kotlin (Kotlin 2.2.0, JVM toolchain 17; AGP 8.5.2)
License	Apache-2.0

Overview

ota-update-engine-bridge is a thin, testable bridge over the AOSP `android.os.UpdateEngine` for native A/B updates. It is intentionally minimal: **no networking, no polling, no business policy, no trust checks** — it accepts an already-verified local (or streaming) artifact descriptor, drives `update_engine.applyPayload(url, offset, size, headers)`, and surfaces typed status. It never writes slot flags, bumps the rollback index, or calls `markBootSuccessful` (`UpdateEngineBridge.kt`, `AndroidUpdateEngineBridge.kt`); `update_verifier` + AVB + the framework take over on the next boot.

The repo splits into two Gradle modules (`settings.gradle.kts`):

- **:core** — a **PURE Kotlin/JVM** library (no Android plugin) holding the framework-independent typed model and raw-int mapping. Fully unit-tested.
- **:android** — the Android-framework wiring. Because `UpdateEngine` / `UpdateEngineCallback` are `@SystemApi` and absent from the public `android.jar`, the implementation binds to them at **runtime via reflection** so the module compiles against the public SDK and runs the real engine only on a platform-signed / system-UID build.

Public API

:core (package digital.vasic.helix.ota.bridge)

- PayloadProperties(fileHashB64, fileSize, metadataHashB64, metadataSize) (PayloadProperties.kt)
 - headersArray(): Array<String> — the 4-element header array in the **canonical, load-bearing order** FILE_HASH, FILE_SIZE, METADATA_HASH, METADATA_SIZE.
 - companion.parse(text): PayloadProperties — parse payload_properties.txt (order-tolerant; rejects missing/duplicate/unexpected keys and non-numeric sizes).
 - Keys object with the four canonical key constants.
- ApplyRequest(url, offset, size, payloadProperties) (ApplyRequest.kt) — validates url non-blank, offset >= 0, size > 0. Helpers: headersArray(), isLocal(file://), isStreaming(http(s)://).
- EngineStatus sealed class (EngineStatus.kt) — typed UpdateEngine status (Idle, CheckingForUpdate, UpdateAvailable, Downloading, Verifying, Finalizing, UpdatedNeedReboot, ReportingErrorEvent, AttemptingRollback, Disabled, plus Unknown(rawCode)); isNeedReboot; companion.fromCode(Int). Raw-int table UpdateStatusConstants.
- EngineError sealed class (EngineError.kt) — typed onPayloadApplicationComplete error (Success, Error, PayloadHashMismatchError, PayloadSizeMismatchError, NotEnoughSpace, DeviceCorrupted, ... plus Unknown(rawCode) for the UNVERIFIED 13..50 range); isSuccess; companion.fromCode(Int). Raw-int table ErrorCodeConstants.

:android (package digital.vasic.helix.ota.bridge.android)

- UpdateEngineBridge interface (UpdateEngineBridge.kt) — the only OS-apply path:
 - applyVerifiedPackage(request: ApplyRequest): ApplyHandle
 - observeStatus(): Flow<EngineEvent> (cold flow; binds on collection, unbinds on cancellation)
 - rebootToNewSlot(handle: ApplyHandle) —
PowerManager.reboot(null) (UpdateEngine has no reboot API).
- EngineEvent sealed interface — Progress(status: EngineStatus, percent: Float), Complete(error: EngineError); ApplyHandle(url).
- AndroidUpdateEngineBridge(context, engine) (AndroidUpdateEngineBridge.kt) — production UpdateEngineBridge over a ReflectiveUpdateEngine.
- ReflectiveUpdateEngine (ReflectiveUpdateEngine.kt) — reflective wrapper exposing applyPayload, bind, unbind, makeCallback, and companion.isAvailable(); RawEngineCallback interface.
- BootStateObserver interface + SystemPropertyBootStateObserver (BootStateObserver.kt) — **read-only** telemetry of currentSlot(), verifiedBootState(), verityMode() via reflective SystemProperties.get. No mutators by design. MergeStatus enum.

Usage

```
import digital.vasic.helix.ota.bridge.ApplyRequest
import digital.vasic.helix.ota.bridge.PayloadProperties
import digital.vasic.helix.ota.bridge.android.AndroidUpdateEngineBridge
import digital.vasic.helix.ota.bridge.android.EngineEvent

// Parse payload_properties.txt and build a request for an
// already-verified local file.
val props = PayloadProperties.parse(
    """
    FILE_HASH=base64sha256...
    FILE_SIZE=1048576
    METADATA_HASH=base64sha256...
    METADATA_SIZE=4096
    """.trimIndent(),
)
val request = ApplyRequest(
    url = "file:///data/ota_package/ota_update.zip",
    offset = 1234L,           // payload.bin offset within the
                           // outer zip
    size = 1048576L,         // payload.bin size
    payloadProperties = props,
)

// On a platform-signed / system-UID build:
val bridge = AndroidUpdateEngineBridge(context)
val handle = bridge.applyVerifiedPackage(request)
scope.launch {
    bridge.observeStatus().collect { event ->
        when (event) {
            is EngineEvent.Progress -> log(event.status,
                event.percent)
            is EngineEvent.Complete -> if (event.error.isSuccess)
                bridge.rebootToNewSlot(handle)
        }
    }
}
```

Testing

The `:core` module is the fully-tested deliverable; run its JVM unit tests:

```
cd submodules/ota-update-engine-bridge
gradle --no-daemon --console=plain :core:test
```

`:core` tests (`core/src/test/.../bridge/`) cover: `ApplyRequest` header delegation + ordering, `file:///http(s)://` classification, and constructor rejection of blank url / negative offset / non-positive size (`ApplyRequestTest.kt`); `EngineStatus` mapping of every known code `0..9`, the verified spec constant table, Unknown-preservation of unpinned codes, and `isNeedReboot` (`EngineStatusTest.kt`); `EngineError` mapping of every known code, the

UNVERIFIED 13..~~50~~ range mapping to Unknown, and isSuccess (EngineErrorTest.kt); and PayloadProperties canonical header order, parse→headersArray round-trip, reordered/blank-line tolerance, and rejection of missing/duplicate/unexpected keys, non-numeric sizes, and malformed lines (PayloadPropertiesTest.kt).

:android is real Kotlin (no stubs), but its build runs the Android Gradle Plugin under the system Gradle — see BUILD_STATUS.md for the honest per-module build/test record in this environment.

Reusable building brick

This is a **reusable, independently versioned** Helix OTA building brick (HelixConstitution §11.4.28 — submodules-as-equal-codebase). Android-only and thin: it wraps the platform API behind a clean, framework-independent-typed interface (:core), so the agent (and any consumer) drives A/B apply without a compile-time @SystemApi dependency. Universal constitution rules are inherited via this repo's CLAUDE.md / AGENTS.md (## INHERITED FROM Helix Constitution).

Mirrors

- GitHub: <https://github.com/HelixDevelopment/ota-update-engine-bridge>
- GitLab: <https://gitlab.com/helixdevelopment1/ota-update-engine-bridge>